

$$M := \frac{A_S^T A_S}{n} - I$$

$$(*) \quad \mathbb{P} \left(\sup_{\|v\|=1} |v^T M v| \geq t \right)$$

$$(**) \quad \mathbb{P} \left(\sup_{v \in \text{Packing}(S^{s-1})} |v^T M v| \geq t \right) \leq 2 \left(\frac{3}{\epsilon} \right)^s e^{-cnt^2} \quad t \leq t_0$$

Goal: To relate (*) & (**) taking advantage of the fact that $\text{Packing}(S^{s-1})$ is an ϵ -approx. of S^{s-1} .

Let $u \in \text{Packing}(S^{s-1})$ s.t. $\|u - v\| \leq \epsilon$.

$$|v^T M v| = |v^T M v + u^T M u - u^T M u|$$

Assuming
that
 $|u^T M u| \leq t$

$$\begin{aligned} &\leq \underbrace{|u^T M u|}_{\substack{u \in \text{Packing} \\ \leq t}} + \underbrace{|v^T M v - u^T M u|}_{\substack{(v-u)^T M v - (u-v)^T M u \\ \leq |v-u|^T M v + |u-v|^T M u \\ \leq \|u-v\|_2 \|Mv\|_2 \\ \leq \epsilon \|M\|_2}} \\ &\leq t + 2\epsilon \|M\|_2 \end{aligned}$$

$$|v^T M v| \leq t + 2\epsilon \|M\|_2$$

$$\sup_{v \in S^{s-1}} |v^T M v| \leq t + 2\epsilon \|M\|_2$$

$$\underbrace{\sup_{v \in S^{s-1}} |v^T M v|}_{\|M\|_2}$$

$$\Rightarrow (1-2\epsilon) \|M\|_2 \leq t$$

$$\Rightarrow \|M\|_2 \leq \frac{t}{1-2\epsilon}$$

Let $\epsilon = 1/4$. $\|M\|_2 \leq 2t$

So, we showed that:

$$\sup_{v \in \text{Packing}(S^{s-1})} |v^T M v| \leq t \Rightarrow \sup_{v \in S^{s-1}} |v^T M v| \leq 2t$$

$\epsilon = 1/4$

$$\mathbb{P} \left(\sup_{v \in S^{s-1}} |v^T M v| \geq \underbrace{2t}_{t'} \right) \leq 2 \cdot \left(\frac{3}{4} \right)^s \cdot \underbrace{e^{-c' n t'^2}}_{e^{-c' n t'^2}}$$

$t = t'/2$

Recall that $\text{RIP}(A, s, \delta)$ requires that

$$\forall S, |S| \leq s, \quad \|A_S^T A_S / n - I\|_2 \leq \delta.$$

$$\mathbb{P} \left(\exists S, |S| \leq s, \sup_{v \in S^{s-1}} |v^T M_S v| \geq t \right)$$

$$\leq 2 \left[\sum_{i=1}^s \binom{d}{i} \right] 12^s e^{-c' n t^2}$$

$$\leq 2s \binom{d}{s} 12^s e^{-c' n t^2}$$

$$\leq \left(\frac{e d}{s} \right)^s$$

$$\leq 2s \left(\frac{12 e d}{s} \right)^s e^{-c' n t^2}$$

$$\leq \delta$$

$$t = 1/3$$

$$C + \log s + s \log(12 e d / s) - c' n t^2 \leq \log \delta$$

$$\Rightarrow n \geq G \left(s \log(12 e d / s) + \log(1/\delta) \right)$$

$$t \leq t.$$

Side Note:

$$\left(\frac{d}{s} \right)^s \leq \binom{d}{s} \leq \left(\frac{d \cdot e}{s} \right)^s$$

$$\binom{d}{s} = \frac{d!}{s!(d-s)!}$$

$$\Rightarrow \frac{d(d-1) \cdots (d-s+1)}{s(s-1) \cdots 1} \geq \left(\frac{d}{s} \right)^s$$

$$\frac{d-i}{s-i} \geq \frac{d}{s}$$

$$d-s-i \geq d-s-d_i$$

$$e^s = \sum_{k=0}^{\infty} \frac{s^k}{k!} \geq \frac{s^s}{s!}$$

$$\Rightarrow \frac{1}{s!} \leq \frac{e^s}{s^s}$$

$$\frac{d(d-1)\dots(d-s+1)}{s!} \leq \underbrace{d^s \cdot \frac{e^s}{s^s}}_{\left(\frac{ed}{s}\right)^s}$$

□

Chapter 15 : Lower Bounds on Estimation

$$X_1, \dots, X_n \stackrel{iid}{\sim} P_{\theta}$$

$$\hat{\theta} = g(X_1, \dots, X_n)$$

$$\min_{\hat{\theta}} \mathbb{E} \{ \|\hat{\theta} - \theta\|_2 \} \quad \|\cdot\|_2 \rightarrow \rho$$

Φ increasing

$$\inf_{\hat{\theta}} \mathbb{E}_{P_{\theta}} \{ \Phi(\rho(\hat{\theta}, \theta)) \}$$

$\rho(\cdot, \cdot)$ distance metric

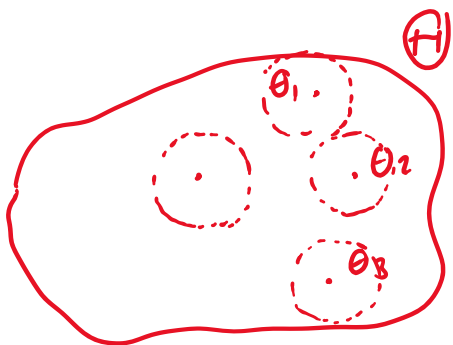
$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E} \{ \underbrace{\Phi(\rho(\hat{\theta}, \theta))}_Z \} \geq m$$

Markov Ineq. $\mathbb{P}(Z \geq t) \leq \frac{\mathbb{E}(Z)}{t} \quad Z \geq 0$

$$\mathbb{E}(Z) \geq \underbrace{t}_{\Phi(\delta)} \cdot \mathbb{P}(Z \geq t) \quad \Phi(\delta)$$

$$\rightarrow \sup_{\theta \in \Theta} \mathbb{E} \{ \Phi(\rho(\hat{\theta}, \theta)) \} \geq \underbrace{\widehat{t}}_{\Phi(\delta)} \cdot \mathbb{P}(\underbrace{\Phi(\rho(\hat{\theta}, \theta))}_{\geq \widehat{t}})$$

$$\sup_{\theta \in \Theta} \mathbb{P}(\rho(\hat{\theta}, \theta) \geq \delta)$$



↓
put lower bounds on it

2δ -separated set

$$\{\theta_1, \dots, \theta_M\}$$

$$\sup_{\theta \in \mathcal{H}} \mathbb{P}(\rho(\hat{\theta}, \theta) \geq \delta) \geq \underbrace{\frac{1}{M} \sum_{i=1}^M \mathbb{P}_{\theta_i}(\rho(\hat{\theta}, \theta_i) \geq t)}$$

$$\mathcal{Q} \quad J \sim \text{Unif} \{1, \dots, M\}$$

$$Z \sim \mathbb{P}_{\theta_J}$$

$$\mathcal{Q}(\rho(\hat{\theta}, \theta_i) \geq t)$$